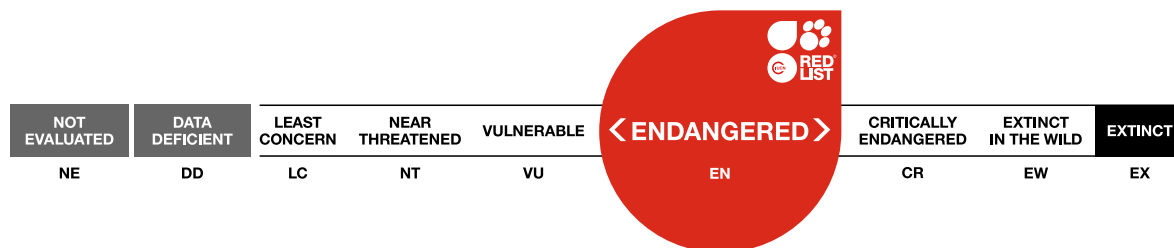


Heleophryne rosei, Table Mountain Ghost Frog

Assessment by: Measey, G.J., Tolley, K.A., Turner, A.A. & Weeber, J.



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Taxonomy

Kingdom	Phylum	Class	Order	Family
Animalia	Chordata	Amphibia	Anura	Heleophrynidae

Scientific Name: *Heleophryne rosei* Hewitt, 1925

Common Name(s):

- English: Table Mountain Ghost Frog
- Afrikaans: Tafelberg-spookpadda
- Xhosa: elibuthunzelarha leNtaba yeTafile

Taxonomic Source(s):

Frost, D.R. 2025. Amphibian Species of the World: an Online Reference. Version 6.2. New York, USA. American Museum of Natural History. Available at: <https://amphibiansoftheworld.amnh.org/index.php..>

Phaka, F.M., Hugé, J., Ravhuanzwo, F., Vanhove, M.P.M. and du Preez, L.H. 2025. Naming South African frogs and reptiles in nine indigenous languages. *African Journal of Herpetology* 74(1): 87-104.

Taxonomic Notes:

The taxonomy of this genus is in need of revision.

Assessment Information

Red List Category & Criteria: Endangered B1ab(iii) [ver 3.1](#)

Year Published: 2025

Date Assessed: April 9, 2024

Justification:

Assessed as Endangered because even though its extent of occurrence (EOO) is only 23 km², it occurs in two to four threat-defined locations, and there is continuing decline in the extent and quality of its habitat.

Previously Published Red List Assessments

[2017 – Critically Endangered \(CR\)](#)

[2010 – Critically Endangered \(CR\)](#)

2004 – Critically Endangered (CR)

1996 – Vulnerable (VU)

1994 – Endangered (E)

Geographic Range

Range Description:

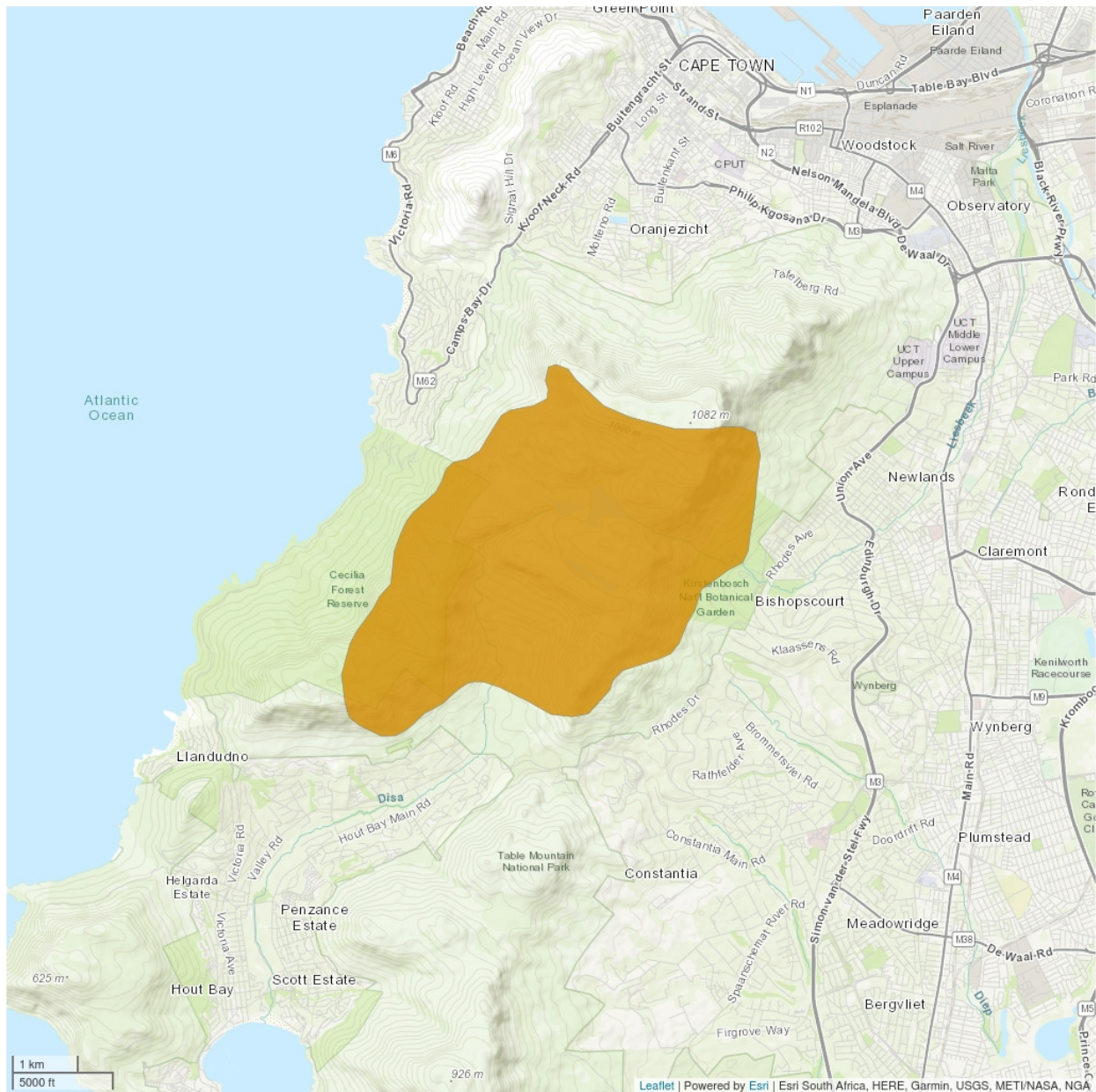
This species has a very restricted range and is endemic to the southern and eastern slopes of Table Mountain, in the Western Cape province, extreme southwestern South Africa, between 240–1,060 m asl. It was historically recorded from seven streams but is currently known to occur in only five streams that are considered to have a viable number of breeding adults (Skeleton, Windows, Upper Disa, Disa, Original Disa). Recent surveys support previous work showing the species is no longer present in Platteklip Gorge and Cecilia streams (Weeber *et al.* 2023), confirming that it has been locally extirpated there for decades (Minter *et al.* 2004, Weeber *et al.* 2023). Its extent of occurrence (EOO) is 23 km². In addition, there are four streams (Valley of the Red Gods, Nursery, Myburgh's Waterfall Ravine, and Hiddingh Ravine) in which tadpoles occur in very low densities in sub-optimal habitat, and breeding is likely sporadic (i.e. not every year). These streams have lower flow and are likely prone to drying up during particularly low rainfall years. They are considered sink populations. Historical records from Valley of the Red Gods probably refer to upper Disa records.

The location and source of known threats are diverse and the species is therefore considered to occur at two to four threat-defined locations.

Country Occurrence:

Native, Extant (resident): South Africa (Western Cape)

Distribution Map



Legend

■ EXTANT (RESIDENT)

Compiled by:

IUCN (International Union for Conservation of Nature) 2024



The boundaries and names shown and the designations used on this map do not imply any official endorsement, acceptance or opinion by IUCN.

Population

Adults of this species are elusive, and are only seen occasionally, whereas tadpoles are typically plentiful where habitat is suitable. The global population is estimated to be stable although the number of tadpoles at impacted sites (e.g. Skeleton Gorge as a result of sedimentation and Original Disa due to water abstraction) may have decreased over the last 50 years (Weeber *et al.* 2023). Tadpoles were historically found in Cecilia Gorge, but they have not been recorded since 1980 (Minter *et al.* 2004, Ebrahim *et al.* 2020, Weeber *et al.* 2023) due to heavy land transformation and afforestation around that stream. It is thought that they are locally extinct at that locality - despite annual occupancy surveys at Cecilia tadpoles have not been rediscovered in the stream (Weeber *et al.* 2023). Genetically there is one population with no differences between streams (J. Weeber unpubl. data April 2024) indicating that adults are capable of moving between streams throughout the mountain. Upper Disa and Disa streams appear to contain the majority of the individuals in the global population for tadpoles and mature adults.

Current Population Trend: Stable

Habitat and Ecology (see Appendix for additional information)

Adults of this species generally occur in both forest and fynbos biomes, breeding in clear perennial headwater mountain streams in ravines on Table Mountain. Adults are found in damp, sheltered habitat next to streams and have also been found well away from streams, including in caves and under rocks. Adults typically select elevated, inaccessible rock cracks. Tadpoles have a large oral disc that allows them to adhere to rocks, streamlined bodies and a wedge-shaped head enabling them to inhabit steeply sloped, fast flowing streams. They rely on sediment free pools with low embeddedness and an abundance of large substrates that provide pore-space refugia (i.e. gaps under/between pebbles and boulders). Tadpoles utilize these spaces to hide and feed, and retreat to them during periods of high winter flow. Tadpoles require longer than 12 months to complete metamorphosis (Weeber *et al.* 2023), and perennial stream flow is therefore important.

Systems: Terrestrial, Freshwater (=Inland waters)

Use and Trade (see Appendix for additional information)

There are no records of this species being utilised.

Threats (see Appendix for additional information)

The main threats are the spread of alien vegetation, the presence of water storage reservoirs on the mountain which can affect the consistency of stream flow, and sedimentation as a result of localized soil erosion from footpaths.

Plantations (for exotic trees) used to be a threat but have now been largely removed so that these no longer pose a primary risk to the species. Varying densities of *Pinus pinaster* and *Populus x canescens* continue to spread in parts of the species range and are still impacting habitat. However, the primary alien plant threatened tadpole habitat is *Acacia melanoxylon*, which has replaced indigenous riparian habitat in Disa Stream.

Water abstraction by the City of Cape Town from Disa and Original Disa streams continues to put pressure on natural hydrological cycles, particularly during early summer when the first rains of winter are caught by the dams and stream flow is reduced. Significant sections of tadpole habitat directly below these dams, and also directly below abstraction weirs situated along the streams length, have been compromised and do not offer appropriate habitat for tadpoles (J. Weeber pers obs. 2023). Fortunately, during winter there appears to be sufficient natural gorge flow to reduce the impacts of abstraction. Two additional weirs on Nursery and Windows streams are managed by Kirstenbosch National Botanical Gardens. Irresponsible abstraction has reduced tadpole habitat directly below these weirs, but recent collaborative work suggests abstraction is now managed appropriately.

Soil erosion caused by human pedestrian traffic on hiking paths is an important threat that has likely caused localized tadpole extinctions in several sections of stream. Increased sedimentation reduces the available pore-space refugia (crucial tadpole microhabitat) and tadpoles are subsequently excluded from these areas.

Climate change is considered a plausible future threat. This species relies on cold, perennial mountain streams and occurs in the upper zones of headwater mountain streams. Reduced rainfall and increased temperatures are therefore real threats. Intense storm events that currently characterize winter storms are also thought to shape tadpole habitat, flooding streams and flushing out excess sediments. A reduction in the intensity and interval between high rainfall events would decrease the abundance of tadpole microhabitat. The species may experience climatic niche displacement under future climate change.

Conservation Actions (see Appendix for additional information)

Conservation Actions In-Place

Most of the range is incorporated in the Table Mountain National Park, which is believed to protect more than 99% of the species' overall distribution, with the rest on the estate of Kirstenbosch National Botanic Gardens (South African National Biodiversity Institute) which has no special protection. The species has been considered Endangered since 1998 (and Critically Endangered since 2004). Since 2019 the species was in an intensive monitoring programme that assessed tadpole distribution across the species range and modelled tadpole presence against environmental covariates, linking results to threat locations and impacts. Research and monitoring are being carried out to estimate population size through occupancy modelling, and to establish whether there is connectivity between streams using genetic analyses. Each stream is also being systematically surveyed over their entire lengths to record presence/absences. The monitoring of tadpole populations and threats to stream habitat will continue.

Conservation Needed

Hiking paths on Table Mountain need to be actively managed to mitigate erosion and localised habitat degradation. This involves the installation of boardwalks in areas where paths bisect streams, stabilising paths that run alongside stream banks, the permanent closure of paths that are routed directly within the stream channel and potentially limiting the numbers of users. Alien vegetation, particularly species such as Deciduous Grey Poplars (*Populus x canescens*) and Australian Blackwood (*Acacia melanoxylon*) that grow in the stream channel, should be removed, and monitored for follow-up annually. It is recommended that water abstraction from weirs situated in reservoir-fed streams is limited to the winter months of May through August, when high run-off and discharge buffers the impact of abstraction. Reservoirs and weirs should be left open outside this suggested time window, specifically

during late summer to ensure that the first wet season rain drains naturally into streams and is not caught by reservoirs and weirs. Rehabilitation actions should focus on Cecilia Gorge.

Research Needed

Tadpole monitoring should continue using the methods outlined in Weeber *et al.* (2023). Detailed, long-term discharge estimates should be recorded for all streams to build a baseline of hydrological data. This will enable researchers to monitor changes in flow and calculate current ecological reserves to inform stream management and abstraction. Acoustic monitoring of breeding adults would assist with tadpole monitoring and could also be used to estimate the population of breeding adults.

Credits

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**Facilitator(s) and
Compiler(s):** Tolley, K.A. & Weeber, J.

Authority/Authorities: IUCN SSC Amphibian Specialist Group

Bibliography

Ebrahim, Z., de Villiers, A. and Measey, J. 2020. Assessing water conditions for *Heleophryne rosei* tadpoles and the conservation relevance. *Koedoe* 62(1): a1581.

IUCN. 2025. The IUCN Red List of Threatened Species. Version 2025-2. Available at: www.iucnredlist.org. (Accessed: 10 October 2025).

Minter, L.R., Burger, M., Harrison, J.A., Braack, H.H., Bishop, P.J. and Knoepfer, D. 2004. *Atlas and Red Data Book of the Frogs of South Africa, Lesotho and Swaziland*. SI/MAB Series No. 9, Washington, D.C.

Weeber, J., Altwegg, R., Tarrant, J. and Tolley, K.A. 2023. Fine-scale drivers of extinction risk: tadpole occupancy dynamics of the Table Mountain Ghost Frog (*Heleophryne rosei*). *African Zoology* 58(3-4): 106-119.

Citation

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External Resources

For [Supplementary Material](#), and for [Images and External Links to Additional Information](#), please see the Red List website.

Appendix

Habitats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Habitat	Season	Suitability	Major Importance?
1. Forest -> 1.4. Forest - Temperate	Resident	Suitable	Yes
3. Shrubland -> 3.8. Shrubland - Mediterranean-type Shrubby Vegetation	Resident	Suitable	-
5. Wetlands (inland) -> 5.1. Wetlands (inland) - Permanent Rivers/Streams/Creeks (includes waterfalls)	Resident	Suitable	-
7. Caves and Subterranean Habitats (non-aquatic) -> 7.1. Caves and Subterranean Habitats (non-aquatic) - Caves	Resident	Suitable	-

Threats

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Threat	Timing	Scope	Severity
2. Agriculture & aquaculture -> 2.2. Wood & pulp plantations -> 2.2.2. Agro-industry plantations	Past, unlikely to return	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	
6. Human intrusions & disturbance -> 6.1. Recreational activities	Ongoing	Majority (50-90%)	Slow, significant declines
	Stresses:	2. Species Stresses -> 2.2. Species disturbance	
7. Natural system modifications -> 7.2. Dams & water management/use -> 7.2.11. Dams (size unknown)	Ongoing	Majority (50-90%)	Slow, significant declines
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.1. Unspecified species	Ongoing	Minority (<50%)	Unknown
	Stresses:	1. Ecosystem stresses -> 1.2. Ecosystem degradation	
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Pinus pinaster)	Ongoing	Majority (50-90%)	Unknown
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion	
8. Invasive and other problematic species, genes & diseases -> 8.1. Invasive non-native/alien species/diseases -> 8.1.2. Named species (Acacia melanoxylon)	Ongoing	Majority (50-90%)	Unknown
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion	
11. Climate change & severe weather -> 11.1. Habitat shifting & alteration	Future	Unknown	Unknown
	Stresses:	1. Ecosystem stresses -> 1.1. Ecosystem conversion 1. Ecosystem stresses -> 1.2. Ecosystem degradation	

11. Climate change & severe weather -> 11.2. Droughts	Future	Unknown	Unknown
Stresses: 1. Ecosystem stresses -> 1.1. Ecosystem conversion			
11. Climate change & severe weather -> 11.3. Temperature extremes	Future	Unknown	Unknown
Stresses: 1. Ecosystem stresses -> 1.1. Ecosystem conversion			

Conservation Actions in Place

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action in Place
In-place land/water protection
Percentage of population protected by PAs: 99
Occurs in at least one protected area: Yes

Conservation Actions Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Conservation Action Needed	Notes
2. Land/water management -> 2.1. Site/area management	-
2. Land/water management -> 2.2. Invasive/problematic species control	-
2. Land/water management -> 2.3. Habitat & natural process restoration	-

Research Needed

(<http://www.iucnredlist.org/technical-documents/classification-schemes>)

Research Needed	Notes
1. Research -> 1.5. Threats	-
3. Monitoring -> 3.1. Population trends	-
3. Monitoring -> 3.4. Habitat trends	-

Additional Data Fields

Distribution
Continuing decline in area of occupancy (AOO): Unknown
Estimated extent of occurrence (EOO) (km ²): 23
Continuing decline in extent of occurrence (EOO): No
Number of Locations: 2-4

Distribution
Continuing decline in number of locations: No
Extreme fluctuations in the number of locations: No
Lower elevation limit (m): 240
Upper elevation limit (m): 1,060
Population
Population severely fragmented: No
Habitats and Ecology
Continuing decline in area, extent and/or quality of habitat: Yes
Movement patterns: Not a Migrant

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